

LESSON 8.7

VOLUME OF RECTANGULAR PRISMS



LESSON INTRODUCTION

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

LEARNING TARGETS

- I can find the volume of a rectangular prism.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- How can you find the volume of a rectangular prism?
- How is the method for calculating the area of a rectangle related to the method for calculating the volume of a rectangular prism?

LESSON NARRATIVE

This lesson builds on students' experiences in packing rectangular prisms with cubes whose side lengths have rational values and introduces problem-solving with volume in real-world contexts. Students compare the two methods for calculating volume and discover a new volume formula ($V=Bh$), then explore real-world application problems with volume, including problems where smaller cube units are encased within a rectangular prism.

COMPONENT SUMMARY

8.7.1 WARM-UP (~5 MIN.)

Investigate a box of cubes

8.7.3 GUIDED PRACTICE (10-15 MIN.)

Solve multistep problems involving the volume of prisms

8.7.5 WRAP-UP (~5 MIN.)

Assessment on calculating volume

8.7.2 GUIDED INSTRUCTION (10-15 MIN.)

Explore and review two methods to calculate volume of rectangular prisms

8.7.4 YOUR TURN! (5 MIN.)

Independent practice on solving real-world volume problems

RESOURCES

GLOSSARY

Key Terms:

- Volume

Additional Terms:

- Unit cube
- Dimension, Base

MATERIALS

- (Optional) Chart paper
- (Optional) Markers
- (Optional) Unit cube or rectangular prism manipulatives
- (Optional) Four-function calculator

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with visualizing the number of cubes packed within a prism and keeping track of the different values used throughout the problems.
- Students may struggle with operational fluency with fractions and mixed numbers.

THINGS TO CONSIDER

- Virtual applets like Desmos can help students to visualize filling a rectangular prism with fraction unit cubes.
- Consider posting an anchor chart of fraction operations for students to reference throughout the lesson.
- Allowing students to use calculators during the lesson will help students focus on discovery and process of finding volume of a rectangular prism. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with positive decimals and positive fractions. For example, question 1 in 8.7.2 Guided Instruction could be a no calculator problem but question 2 in 8.7.2 Guided Instruction could be a calculator problem.

LITERACY AND LANGUAGE ROUTINES

Number Talk

8.7.3 Guided Practice provides an opportunity to build computational fluency by encouraging students to compare different fractional unit lengths within a prism. Consider using Number Talk routines to facilitate number sense and reasoning related to finding volume with fractional side lengths as factors.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Students will be working to solve real-world problems where they must find the total number of cube units that can be packed within a larger rectangular prism. Consider encouraging students to predict solutions before solving the problem.

DIFFERENTIATE INSTRUCTION

In 8.7.3 Guided Practice, consider providing diagrams of blank rectangular prisms for students to use to help with questions 1 and 2. Students can mark up the sides of the prism with the given fractional units to assist them in finding the total number of cubes enclosed in a prism. Isometric dot paper is also a helpful support if students would like to draw the prisms from scratch.

To foster kinesthetic reasoning about volume with fractional side lengths, provide unit cubes or similar math manipulatives for students to build prisms representing the problems in 8.7.4 Your Turn! or 8.7.5 Wrap-Up sections.

8.7.1 WARM-UP: A BOX OF CUBES

Component Overview: Students investigate volume by finding the total number of unit cubes inside a rectangular prism, recalling the previous lesson's activities. Students then determine the total number of cubes that can fit within the prism if the side lengths are increased or decreased.

TEACHER MOVES

Consider positioning students to discuss question prompts rather than writing their answers. This auditory entry point might engage learners and foster correct use of math vocabulary and problem-solving strategies. Use sentence stems to encourage thinking such as:

- “I had to pay attention when problem-solving when...”
- “One detail that I have to remember a lot in math class is...”

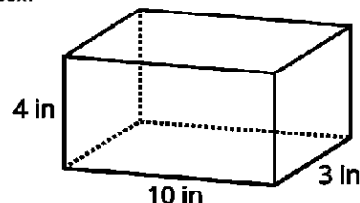
DIFFERENTIATE INSTRUCTION

Consider prompting students to draw a sketch for both questions 2 and 3. This can help students visually make the connection between how the edge length affects how many cubes can fit into the box. It may be helpful to also have a physical manipulative to share whole group (comparing 1 unit to 2 units to 1/2 unit).



8.7.1 Warm-Up: A Box of Cubes

1. How many cubes with an edge length of 1 inch (in.) fill the box?



$$V_{\text{Box}} = lwh = 4 \text{ in} \times 10 \text{ in} \times 3 \text{ in} = 120 \text{ in}^3$$

$$V_{\text{Cube}} = lwh = 1 \text{ in} \times 1 \text{ in} \times 1 \text{ in} = 1 \text{ in}^3$$

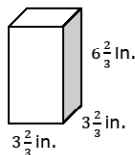
$$\text{Number of cubes} = \frac{120 \text{ in}^3}{1 \text{ in}^3} = 120 \text{ cubes}$$

2. If the cubes had an edge length of 2 inches, explain if more or fewer cubes would be needed to fill the box.
Fewer cubes would be needed because the cubes would be bigger and fill more space.
3. If the cubes had an edge length of $\frac{1}{2}$ inch, explain if more or fewer cubes would be needed to fill the box.
More cubes would be needed to fill the box because the cubes would be smaller and take up less space.



8.7.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism

Alan calculated the volume of the rectangular prism.



- Fill in the blanks to complete Alan's statement on how he calculated the volume.

To find the volume of this prism, I will fill the prism with cubes of side lengths $\frac{1}{3}$ in. Because $3\frac{2}{3} \div \frac{1}{3} = \underline{11}$, 11 cubes will fit along the length and 11 cubes will fit along the width. 121 cubes will cover the base of the prism. The height of the prism can be used to determine how many layers of cubes are needed. It will take 20 layers of cubes because $6\frac{2}{3} \div \frac{1}{3} = \underline{20}$. In total, it will take 2420 of the $\frac{1}{3}$ in. cubes to fill the prism.

The volume of each small cube is $\frac{1}{27}$ in³.

- Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism.
 $89\frac{17}{27}$ in³

- Jai used the formula $V = l \times w \times h$ to find the volume of the prism. Complete Jai's work to find the volume of the prism.

$$V = l \times w \times h$$

$$V = 3\frac{2}{3} \times 3\frac{2}{3} \times 6\frac{2}{3}$$

$$V = 89\frac{17}{27} \text{ in}^3$$

- How are Alan and Jai's methods similar?
Answers will vary. Alan and Jai's methods were similar because they both multiplied the sides together.



The volume, V , of a rectangular prism is

$V = l \times w \times h$, where l represents the length, w represents the width, and h represents the height. This can also be written as $V = B \times h$, where B represents the area of the base and h represents the height.

- Discuss with your partner how the formula $V = B \cdot h$ is related to the formula $V = l \cdot w \cdot h$.
The area of the base (B) is found by multiplying length times width.
- Complete the statements based on your discussion.

The product of the length, l , and the width, w , is equal to

the area of the base of the prism.

$$B = l \times w$$

Both formulas can be used to find the volume of a right rectangular prism.

8.7.2 GUIDED INSTRUCTION: TWO METHODS FOR FINDING THE VOLUME OF A RIGHT RECTANGULAR PRISM

Component Overview: Students apply strategies to find the volume of right rectangular prisms. Question 1 involves a guided review on how to calculate the volume of the prism with fractional edges by packing cubes and by applying the volume formula ($V = lwh$). The second section leads students to discover that another formula ($V = Bh$) can be used to calculate the volume of a right rectangular prism.

TEACHER MOVES

Consider which questions students should use a calculator and which questions provide good practice for fluency.

ENRICHMENT

Is there another way to find the volume of this prism besides using these small cubes? Which strategy do you prefer?

ENRICHMENT

- What is considered the "base" of a rectangular prism?
- If we rotate the prism, does the base change? Will the formula change?

TEACHER MOVES

Specifically have students make connections between $B = l \times w$ and the shape of the base in a rectangular prism. Since the base is a rectangle, the area formula used for B reflects the area of a rectangle ($A = l \times w$).

8.7.3 GUIDED PRACTICE: VOLUME OF A RIGHT RECTANGULAR PRISM

Component Overview: Students transition from teacher-led 8.7.2 Guided Instruction to student-led 8.7.3 Guided Practice. In this activity, students are prompted to both contemplate and calculate in order to have opportunity to make sense of information while finding solutions.

DIFFERENTIATE INSTRUCTION

Consider providing a diagram of a rectangular prism to manipulate for these exercises. This visual and kinesthetic support can help students better understand how to use given information to find volume of various prisms.

ENRICHMENT

To extend student thinking and reasoning:

- Does it matter which fractional unit cubes we use to find the volume?
- Could we use a different fractional unit cube for these examples?

TEACHER MOVES

Number Talk

Question 2 provides students with the opportunity to apply different fractional unit cubes to find the volume of a prism. Consider encouraging partner pairs to calculate the volume by selecting a different fractional unit cube. They should confirm with one another that they have arrived at the same volume and see if they notice a pattern between the numbers they used.



8.7.3 Guided Practice: Volume of a Right Rectangular Prism

- Diego correctly points out that 81 cubes with an edge length of $\frac{1}{3}$ centimeters (cm) are needed to fill a right rectangular prism that has length of 3 cm, width of 1 cm, and height of 1 cm.
 - Explain or show how this is true.
81 of the cubes with a volume of $\frac{1}{27} \text{ cm}^3$ would have a combined volume of 3 cm^3 which matches the volume when multiplying $l \cdot w \cdot h$ of the right rectangular prism.
 - What is the volume, in cubic centimeters, of the rectangular prism?
 3 cm^3
- Lin and Noah are packing small cubes into a cube with an edge length of $1\frac{1}{2}$ inches. Lin is using cubes with an edge length of $\frac{1}{2}$ inch, and Noah is using cubes with an edge length of $\frac{1}{4}$ inch.
 - Explain who would need more cubes to fill the $1\frac{1}{2}$ in. cube.
Noah would need more cubes because $\frac{1}{4}$ in. cubes are smaller and fill less space than $\frac{1}{2}$ in. cubes.
 - Explain whether Lin and Noah will find the same volume of the $1\frac{1}{2}$ inch cube using their method.
They will find the same volume using their own method because even though Noah will need more cubes, his cubes are smaller and will fill up the same amount of space.

- c. What is the volume of the cube with $1\frac{1}{2}$ inch edge lengths? Include units.

$$3\frac{3}{8} \text{ in}^3$$

3. A toy company is packaging its toys to be shipped. Some small toys are placed inside a cube-shaped box with side lengths of 0.5 inches (in.). These smaller boxes are then packed into a shipping box with dimensions of 12 in. $\times$ 35 in. $\times$ 2.5 in.

- a. How many small toys can be packed into the larger box for shipping?

8,400 small toys

- b. Use the number of toys that can be shipped in the box to determine the volume of the box. Include units.

1,050 in³

8.7.3 GUIDED PRACTICE: VOLUME OF A RIGHT RECTANGULAR PRISM (CONTINUED)

ENRICHMENT

To scaffold thinking for question 3, consider asking:

- How many cube-shaped boxes will fit into the length of the shipping box? Width? Height?
- For Part 8a, does this answer seem reasonable? How do you know?

DIFFERENTIATE INSTRUCTION

Consider assigning question 1 or question 2, but not both, for students who require additional processing time to complete arithmetic work and focus on the conceptual and procedural understanding required to find volume.

QUESTIONING

For early finishers, consider an additional challenge by changing the size of the toys. Students could first make a guess for how many 1-inch or 2.5-inch-size cube boxes could fit and then calculate to test their hypotheses.

8.7.4 YOUR TURN!

Component Overview: Students engage with real-world problems involving volume. Students should independently make sense of problems and apply methods of finding volume to practice in real-world contexts.

TEACHER MOVES

Consider asking students what they know about salt licks. Allowing students to share something they have knowledge of that other students may not helps to build their confidence.

Review strategies to find the total number of salt cubes and strategies to determine the dimensions of the cube before moving on to the 8.7.5 Wrap-Up. Possible strategies include:

- Applying the volume formula to the side lengths and then dividing by the volume of the salt cubes.
- Dividing each dimension by $\frac{5}{12}$ and then calculating the total number of cubes.

Use this as an opportunity to clarify any lingering misconceptions.

QUESTIONING

Consider the entry point for question 1a and which students might benefit most from engaging with this question prompt. For additional complexity, have students attempt to provide counterexamples of when this might not be true.

ENRICHMENT

What strategy for calculating the volume is most efficient for question 2? Why?



8.7.4 Your Turn!

- To give their animals essential minerals and nutrients, farmers and ranchers often have a block of salt, called a "salt lick" available for their animals. A rancher is ordering a box of cube-shaped salt licks. The edge lengths of each salt lick are $\frac{5}{12}$ foot (ft).



- Explain if the volume of one salt lick is greater than or less than 1 cubic foot (ft^3).

The volume of one salt lick is less than 1 cubic foot because when you multiply the sides, which are fractions, your resulting product is smaller.

- A box that is $1\frac{1}{4}$ ft $\times$ $1\frac{2}{3}$ ft $\times$ $1\frac{5}{6}$ ft is used to ship the salt licks. How many cubes of salt lick can fit in the box?

$1\frac{5}{12} \div \frac{5}{12} = 3$ fit along this dimension of the box

$1\frac{2}{3} \div \frac{5}{12} = 4$ fit along this dimension of the box

*$1\frac{5}{6} \div \frac{5}{12} = 4.4$, so 4 fit along this dimension of the box
 $3 \times 4 \times 4 = 48$ salt licks fit in the box*

Some students may not recognize they can't cut salt licks to fit in the box, so may simply find the volume of each and divide to find 52.8 salt licks. This shows understanding of volume, but not the context.

- A pool in the shape of a rectangular prism is being filled with water. The dimensions of the pool are 24 feet (ft) by 15 ft. If the height of the water in the pool is 1.3 ft, what is the volume of the water in cubic feet?

$V = lwh = 24\text{ft} \times 15\text{ft} \times 1.3\text{ft} = 468\text{ft}^3$



8.7.5 Wrap-Up: Ticket Out the Door

1. Joy keeps her modeling clay in a plastic box that has a length of 21 centimeters (cm), a width of 21 cm, and a height of 63 cm.

- a. What is the volume of the plastic box? Include units.

$$V_{\text{plastic box}} = 21 \text{ cm} \times 21 \text{ cm} \times 63 \text{ cm} = 27,783 \text{ cm}^3$$

- b. If each piece of cube-shaped clay has a side length of 4.2 cm, how many pieces of clay can be stored in the box?

$$V_{\text{clay}} = lwh = 4.2 \text{ cm} \times 4.2 \text{ cm} \times 4.2 \text{ cm} = 74.088 \text{ cm}^3$$

$$N_{\text{clay boxes}} = \frac{27,783 \text{ cm}^3}{74.088 \text{ cm}^3} = 375 \text{ boxes}$$

8.7.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to calculate the volume of a right rectangular prism and to find a total number of cubes encased within the prism.

DIFFERENTIATE INSTRUCTION

For students who might struggle with math organization in multistep problems, consider providing scaffolded steps for students to fill in to demonstrate their understanding.

An example is:

- a. $V = lwh$

$$V = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

$$V = \underline{\hspace{2cm}}$$

- b. 4.2 cm cubes fit in the length

 4.2 cm cubes fit in the width

 4.2 cm cubes fit in the height

 total 4.2 cm cubes